WORKED SOLUTION

STEP II 1999, Question 13

Probability and Statistics • Continuous Random Variables

STEP II 1999, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q13

The key idea

Although it is filed under mechanics, this is a question about a continuous random variable. Everything becomes easy once you notice that the *shorter* piece has a simple distribution: if the break point is uniform along the stick, the shorter piece is uniform on $[0, \frac{1}{2}]$ of the length. The ratio R is an increasing function of the shorter piece, so "$R < r$" turns into a statement about the shorter piece alone. That gives the distribution function, and differentiating gives the density.

Solution

The distribution of R

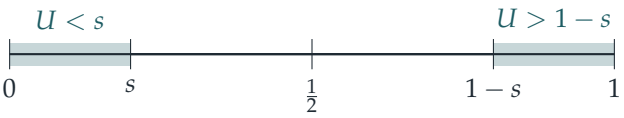
Take the stick to have length 1; ratios don't depend on the units. Let the break be at distance U from one end, where U is uniformly distributed on $[0, 1]$. The two pieces have lengths U and $1 - U$, so the shorter one has length

$$S = \min(U, 1 - U),$$

and the longer one has length $1 - S$. Then $0 \leq S \leq \frac{1}{2}$ and

$$R = \frac{S}{1 - S}.$$

For $0 \leq s \leq \frac{1}{2}$, the event $S < s$ happens exactly when $U < s$ or $U > 1 - s$. These are two disjoint intervals of total length $2s$, so $\mathrm{P}(S < s) = 2s$. In other words, S is uniform on $[0, \frac{1}{2}]$.



Now let $0 \le r \le 1$. The function $s \mapsto \dfrac{s}{1-s}$ is strictly increasing on $[0, \frac{1}{2}]$ (its derivative is $1/(1-s)^2 > 0$), so we may invert the inequality:

$$R < r \iff \frac{S}{1-S} < r \iff S < r(1-S) \iff S < \frac{r}{1+r},$$

where multiplying through by $1-S$ is safe because $1-S \ge \frac{1}{2} > 0$. The value $\dfrac{r}{1+r}$ lies in $[0, \frac{1}{2}]$ when $0 \le r \le 1$, so

$$\boxed{\mathrm{P}(R < r) = \frac{2r}{1+r} \qquad (0 \le r \le 1).}$$

Since the shorter piece is never longer than the longer one, R always lies in $[0,1]$. So $\mathrm{P}(R < r) = 0$ for $r \le 0$ and $\mathrm{P}(R < r) = 1$ for $r \ge 1$, which matches the formula's value 1 at $r = 1$.

Ask yourself Why not work with U directly, as $R = U/(1-U)$?

That's only right when $U \le \frac{1}{2}$. When $U > \frac{1}{2}$ the shorter piece is $1-U$, and $R = (1-U)/U$. You can split into these two cases, and each one contributes $\frac{r}{1+r}$, by symmetry. Passing to S does that bookkeeping once and for all.

The density, mean and variance

Differentiating the distribution function $F(r) = \dfrac{2r}{1+r} = 2 - \dfrac{2}{1+r}$ on $(0,1)$ gives the probability density function

$$f(r) = \frac{2}{(1+r)^2} \quad (0 \le r \le 1), \qquad f(r) = 0 \text{ otherwise.}$$

As a check, $\int_0^1 f(r)\,\mathrm{d}r = F(1) - F(0) = 1$.

For the mean, write $r = (1+r) - 1$:

$$\begin{aligned}\mathrm{E}(R) &= \int_0^1 \frac{2r}{(1+r)^2}\,\mathrm{d}r = 2\int_0^1 \left(\frac{1}{1+r} - \frac{1}{(1+r)^2}\right)\mathrm{d}r \\ &= 2\Big[\ln(1+r) + \frac{1}{1+r}\Big]_0^1 = 2\left(\ln 2 + \tfrac{1}{2} - 1\right) \\ &= 2\ln 2 - 1 \approx 0.386.\end{aligned}$$

For the second moment, write $r^2 = (1+r)^2 - 2(1+r) + 1$, so that

$$\frac{r^2}{(1+r)^2} = 1 - \frac{2}{1+r} + \frac{1}{(1+r)^2}.$$

Then

$$\begin{aligned}\mathrm{E}(R^2) &= 2\int_0^1 \left(1 - \frac{2}{1+r} + \frac{1}{(1+r)^2}\right)\mathrm{d}r = 2\Big[r - 2\ln(1+r) - \frac{1}{1+r}\Big]_0^1 \\ &= 2\left(1 - 2\ln 2 - \tfrac{1}{2} + 1\right) = 3 - 4\ln 2.\end{aligned}$$

Hence

$$\begin{aligned}\operatorname{Var}(R) &= \mathrm{E}(R^2) - (\mathrm{E}(R))^2 = 3 - 4\ln 2 - (4(\ln 2)^2 - 4\ln 2 + 1)\\ &= 2 - 4(\ln 2)^2 \approx 0.0782.\end{aligned}$$

The $\ln 2$ terms cancel, which leaves this tidy form.

Checks

- At $r = 1$ the probability is 1, as it must be. At $r = \frac{1}{2}$ it is $\frac{2}{3}$: the longer piece is more than twice the shorter exactly when the break lands in the outer thirds of the stick, which have total length $\frac{2}{3}$.
- The mean 0.386 lies between 0 and 1, and below $\frac{1}{2}$, which fits a density that decreases from 2 at $r = 0$ to $\frac{1}{2}$ at $r = 1$.
- A simulation of 10^6 random breaks gave a sample mean of 0.3862 and a sample variance of 0.0783, against $2\ln 2 - 1 = 0.38629$ and $2 - 4(\ln 2)^2 = 0.07819$.

Common slips

- Taking $R = U/(1-U)$ for all U. That quantity can exceed 1, so it isn't the ratio of shorter to longer.
- Getting $\mathrm{P}(R < r) = \frac{r}{1+r}$ by forgetting that the break can be near either end.
- Using the mean of the shorter piece, $\frac{1}{4}$, to get $\mathrm{E}(R) = \frac{1/4}{3/4} = \frac{1}{3}$. In general $\mathrm{E}\big(g(S)\big) \neq g\big(\mathrm{E}(S)\big)$.

Beyond STEP

The method here is the standard "distribution function technique" for transforming a random variable. If $Y = g(X)$ with g increasing, then $F_Y(y) = F_X\big(g^{-1}(y)\big)$, and differentiating gives the change-of-variables formula $f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\mathrm{d}g^{-1}/\mathrm{d}y\right|$. The same idea leads on to order statistics and to the classic broken-stick problems, such as the chance that three pieces form a triangle. Grimmett and Welsh, *Probability: An Introduction*, is a good place to read about transforming continuous random variables.

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